

6	(a) parallel (to the field)	B1	[1]
	(b) (i) torque = $F \times d$ $2.1 \times 10^{-3} = F \times 2.8 \times 10^{-2}$ $F = 0.075 \text{ N}$ <i>(use of 4.5 cm scores no marks)</i>	C1 A1	[2]
	(ii) zero	A1	[1]
	(c) $F = BILN(\sin\theta)$ $0.075 = B \times 0.170 \times 4.5 \times 10^{-2} \times 140$ $B = 7.0 \times 10^{-2} \text{ T} = 70 \text{ mT}$	C1 M1 A0	[2]
	(d) (i) (induced) <u>e.m.f.</u> is proportional to / equal to <u>rate of change</u> of (magnetic) flux (linkage)	M1 A1	[2]
	(ii) change in flux linkage = BAN $= 0.070 \times 4.5 \times 10^{-2} \times 2.8 \times 10^{-2} \times 140$ $= 0.0123 \text{ Wb turns}$ induced e.m.f = $0.0123 / 0.14$ $= 88 \text{ mV}$ <i>(Note: This is a simplified treatment. A full treatment would involve the averaging of $B \cos\theta$ leading to a $\sqrt{2}$ factor)</i>	C1 C1 A1	[3]